SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Users (students and teachers) shall securely log in using a username/email and password to access the system.*	High
2	Authorization levels	*The system shall provide role-based access where students can access learning modules and teachers/admins can monitor progress and analytics.*	High

3	External interfaces	*The system shall integrate with AI/ML APIs (e.g., TensorFlow, OpenCV, MediaPipe, Gemini API) for emotion detection and connect to a cloud database for storing user data.*	High
4	Transactions processing	*The system shall capture user input (video/image), detect emotions in real time, store results, and generate personalized learning recommendations.*	High
5	Reporting	*The system shall generate dashboards showing emotion history, learning progress, engagement levels, and performance reports for students and teachers.*	Medium
6	Business rules, etc.	*The system shall recommend learning resources based on detected emotions (e.g., confusion → simplified content, frustration → motivational support). Only authenticated users can access personalized recommendations.*	High

7	Compliance to laws or regulations	*The system shall protect user privacy by securely storing personal and emotion data, obtaining user consent before capturing images/videos, and complying with applicable data protection regulations.*	High
8	*Other*	*[Add any additional functional requirements here]*	High

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	*The system should detect emotions and display recommendations with minimal delay.*	*Emotion detection response time ≤* ***2 seconds****.*
2	Scalability	*The system should support multiple users simultaneously without performance degradation.*	*Supports* ***500+ concurrent users****.*
3	Security & Data Privacy	*User credentials and emotion data must be securely stored and transmitted.*	*Passwords encrypted, HTTPS enabled, user consent required before accessing camera.*

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**

4	Reliability & Availability	*The system should remain available with minimal downtime.*	***99% uptime** with automatic error recovery.*
5	Usability & Accessibility	*The interface should be simple, responsive, and easy to use on desktop and mobile devices.*	*User can access major features within **3 clicks**; responsive UI.*
6	*Other*	*The system should be easy to maintain, update, and deploy across different platforms.*	Modular codebase, compatible with Windows, Linux, and modern web browsers.